

RISHABH MALVIYA

ML & Cloud Infrastructure Engineer · Production ML Pipelines · Robotics & Computer Vision · MLOps Platforms

San Francisco, CA · U.S. Citizen · +1 (408) 215-1528 · rishabh.malviya@gmail.com

[linkedin.com/in/rish-malviya](https://www.linkedin.com/in/rish-malviya) · github.com/RishabhMalviya · medium.com/@rishabh.malviya

SUMMARY

ML & cloud infrastructure engineer with 8+ years building and deploying high-performance ML pipelines in production. My experience spans robotics, computer vision, fintech, and enterprise SaaS. I turn research models into reliable, monitored, self-improving systems. I also build orchestration tooling, CI/CD, and cloud infrastructure that lets ML teams move fast. I have hands-on experience deploying computer-vision models in harsh real-world environments, and a robotics background (self-driving, SLAM, reinforcement learning).

CORE SKILLS

- **Languages:** Python (10+ yrs, expert), C++, Go (migration experience), Java, Kotlin, TypeScript, Bash, SQL
- **ML / AI:** PyTorch, Computer Vision (YOLOv7, object detection, quantization, TensorRT), Reinforcement Learning (Stanford graduate coursework in RL for Robotics), NLP, Knowledge Graphs (GraphSAGE)
- **ML Pipelines & MLOps:** Cloud training / evaluation / inference pipelines, Kubeflow-like orchestration platform, experiment tracking with MLFlow, automated retraining, shadow testing, model monitoring
- **Cloud & Infra:** AWS (extensive), Azure, Docker, Kubernetes (EKS/ECS, OpenShift), Airflow, Terraform (multi-account IaC), GitHub Actions, GitLab CI/CD, Prometheus & Grafana, GPU provisioning & monitoring, Redis
- **Data:** Scalable data pipelines (1M+ points/day), Snowflake, PostgreSQL (RDS, Supabase), DynamoDB

WORK EXPERIENCE

Machine Learning Lead — Liberata (Duke University)

Feb 2026 – Present

Remote · Python, AWS, Docker, Kubernetes

- Leading the ML team—setting the team’s vision, doing project planning, mentoring, and identifying where ML creates value.
- Building **simulation environments and adversarial RL agents** to surface exploits and emergent behaviors in incentive systems. This work was selected for the national Deloitte-sponsored **AI Venture Velocity Challenge**
- Deliver talks at Duke University to students breaking into ML/AI careers.

Senior Machine Learning Engineer — DocuSign

Aug 2023 – Feb 2026

San Francisco, CA · Python, AWS (SageMaker), Docker, Airflow, Terraform, Snowflake, GitHub Actions, MLFlow, Evidently

- **Project lead** for an in-house, Kubeflow-like **ML orchestration platform** providing training, re-deployment, retraining, monitoring, and tracking for **25+ batch and real-time models**—reducing deployment and maintenance cycles from **2 months to 1 week** for ML models that touch 1000s of employees globally.
- Built **automated retraining pipelines** triggered by continuous production performance and model explainability tracking; drove offline gains of **up to +30 F1 points** and a **20% global sales-conversion uplift**.
- Managed **100+ model & container registries** with automated GitHub Actions life cycle workflows; built developer libraries for cross-environment (DEV/UAT/PROD) data ops plus utilities to **provision and monitor GPU usage**.
- Built the ML monitoring framework using the open-source library Evidently
- Optimized SHAP explain ability pipeline speed by **96%**

Machine Learning (Computer Vision) Engineer — Aquabyte.ai

Apr 2021 – Jan 2023

San Francisco, CA · Python, PyTorch, YOLOv7, AWS, Docker, Kubernetes, Prometheus, Grafana, Go

- Productionized models built with **YOLOv7 (for object detection and keypoint detection)** and ResNet classifiers into a service that automated a 50-person annotation operation, saving **~\$1M annually**, and paving the way for further AI product lines.
- Designed **gradual rollout and shadow testing** strategies to gather unbiased human feedback on deployed models. Orchestrated a safe human-to-AI transition, ensuring successful deployment in **harsh, real-world outdoor environments**.
- Optimized model inference with **TensorRT and quantization**.
- Built systems for tracking business performance metrics and monitoring system metrics with Prometheus & Grafana dashboards.
- Collaborated across distributed teams in Norway, Chile, and SF.
- Re-factored live orchestration services handling **~20 requests per second (each of 100s of MB)**, migrating from **Go + DynamoDB** to a Python microservice + PostgreSQL.

EDUCATION

CS224R - Reinforcement Learning for Robotics (Graduate Course)

Completed June 2026

Stanford University

- Performed research on Body Schema Pretraining (BSP), a framework for pre-training policies on the intrinsic dynamics of a robot embodiment, so that fine-tuning on different downstream tasks can be done faster.

B.Tech, Engineering Physics + Minor in Applied Statistics (GPA 8.1/10)

2013 – 2017

IIT Bombay

- Core member, IIT-B **Self-Driving Car team**: implemented an Extended Kalman Filter for sensor fusion feeding localization and **SLAM** modules (Python & C++/ROS).
- Mentor, IIT-B Innovation Cell — mentored the team that [won the Intelligent Ground Vehicle Challenge 2017](#)
- **Paper published at the IEEE SSCI conference**: [Face & Voice Authentication on Neuromorphic Hardware](#)
- One of the few IIT Bombay undergraduates to secure an international internship at Honda Research Institute - Japan. Developed LSTM-CNN models for Coreference Resolution.

SELECTED PROJECTS (ROBOTICS & DEPLOYMENT)

- [Body Schema Pretraining \(BSP\)](#) — A methodology for RL transfer learning to different locomotion tasks on the same robot embodiment
- [CFD-MLOps](#) - Training a Surrogate Neural Operator Models for CFD Simulations on car meshes using the Transolver architecture with PyTorch DDP
- [ML REST API Deployment Template](#) — a template for packaging and deploying any ML model as a Dockerized REST API with auth and auto-generated docs
- [Siamese-Network Facial Recognition ROS Wrapper](#) — A ROS wrapper for the Python face_recognition library, which implements few-shot learning with Siamese networks
- [Spiking Neural Networks for Arduino](#) — LIF Neurons / Liquid State Machines for Arduino Robots.

SCHOLASTIC ACHIEVEMENTS

- **IIT-JEE 2013**: All-India Rank 490 (top 0.3%) among 150,000 candidates.
- **SAT Subject Tests**: perfect 800/800 in both Mathematics Level II and Physics.
- **NTSE 2008**: National Talent Search Examination scholarship, Government of India.